

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

Hardware and forming

Gear and wheels

G01 is 20 x4 mm 6061-T6 aluminium with a 12 mm inside bend radius. The formed STEP governs final dimensions. $K=0.40$ in the flat DXF is a starting assumption: qualify a bend coupon, correct the allowance for the actual tools, then form and finish-drill. Do not treat the ideal bending calculation as a landing rating.

Nose gear

G02 is 4 mm spring wire, formed by a shop able to preserve the required material properties. A09/A37 bores are 4.2 mm before process adjustment. Fit a 4 mm clamp steering arm and positive shaft retainers. Use the 76.2 mm wheel reference, 22 mm nominal hub and 4 mm bore. The nose axle ends at X46; fit washers/collars without clamping the hub.

Main attachment

Bond G03 below A47. Two M3 through-bolts clamp A47/G03/G01 with broad washers and locking nuts. Use smooth-shank M4 main axles. Select actual bolt lengths for the measured stack and retain two threads beyond the locking nut.

Motor

A26 is 5 mm aircraft plywood. Its four 3.2 mm holes lie on a 19 mm bolt circle, with adjacent centres 13.435 mm apart. Preserve the 12 mm centre opening. Measure M3 screw engagement to avoid windings. The 33.9 x34 mm conservative motor envelope clears the tail fairing; verify the actual bell and leads before running.

Controls

Use five TowerPro MG90S, ten mounting bolts, four purchased lightweight horns, four H05B backing plates and five adjustable pushrods with ten retained rod ends. Align horn pins over hinge axes. Through-fasteners and crush-resistant spacer sleeves carry loads across hollow prints or foam. Set neutral geometry before cutting rods; retain pins and clips mechanically.

Wing pins and retainers

Two smooth 3 mm aileron hinge rods start at 285 mm and are trimmed after fitting retainers, leaving about 1 mm axial freedom. Use three M2 root retainers per wing. Bond front sleeves only; do not glue the sliding joiner into both wings.

Battery and cover hardware

Use the quantities in Complete_Parts_and_Materials.csv. Battery hangers attach to A13; the tray hangs below them. Two straps wrap the pack and tray, with two padded corner stops. Fit nonmagnetic cover/GPS hardware near the compass. Front-stop access must be proved with the chosen screws and tools before covering.

Process checks

Use the plywood slot coupons and 80 x35 x4 mm printer coupon. Current printer-coupon bores are 2.0, 2.2, 3.0, 3.2, 4.0, 4.2, 4.4, 8.0, 8.2 and 10.2 mm. Qualify the actual stock, kerf, printed fit, foam forming and adhesive before making the full set.